

Notes: Khandelwal, Schott, and Wei (AER, 2013)

Trade Liberalization and Embedded Institutional Reform: Evidence from Chinese Exporters

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1 Big picture

- **Question.** How large are the productivity gains from trade liberalization when the trade barrier is administered by a distorted institution? In this paper, the barrier is export quotas under the Multifiber Arrangement (MFA), and the institution is the Chinese government’s quota allocation regime.
- **Setting.** The paper studies Chinese textile and clothing exports to the United States, the European Union, and Canada around the elimination of MFA quotas on January 1, 2005.
- **Core challenge.** Removing a quota does two things at once. It removes the trade barrier itself, but it may also dismantle the domestic licensing institution that was distributing quota rights. The paper asks whether that licensing institution misallocated export opportunities away from the most productive firms.
- **Main theoretical benchmark.** In a Melitz-style model with auction-allocated quota licenses, quotas operate like a per-unit fee. The fee disproportionately hurts low-price, high-productivity firms, so removing quotas should mainly increase the market shares of the most productive incumbents.
- **Main empirical result.** That prediction is strongly rejected in the data. After quota removal, export growth and price declines are driven mainly by net entry rather than incumbents, and market share shifts away from state-owned incumbents toward private entrants.
- **Interpretation.** The Chinese government did not allocate quotas on the basis of firm productivity. Instead, quota allocation appears to have favored relatively inefficient incumbents, especially state-owned enterprises (SOEs), and blocked more productive private firms from entering.
- **Quantitative takeaway.** In the model-based counterfactual, about **71%** of the total productivity gain from removing quotas comes from eliminating the **embedded misallocation in the licensing regime**, while only 29% comes from removing the quota itself.
- **Why this paper matters.** The paper shows that trade liberalization can have larger-than-standard gains because it can also destroy domestic institutions that were distorting allocation. It therefore links trade policy to misallocation, entry, and institutional reform.

2 Data and measurement (key objects)

2.1 Observed exporter-level panel

- **Core customs data.** The main data are Chinese customs transactions by **firm** $\times$ **8-digit HS product** $\times$ **destination country**. For each observation, the paper observes nominal export value, export quantity, and firm ownership type.
- **Ownership categories.** Firms are grouped as **SOEs**, **domestic private firms**, and **foreign-invested private firms**.
- **Prices.** Export prices are nominal FOB unit values, constructed as export value divided by export quantity.

- **Coverage.** Quantity units are observed for about 99% of observations, which makes plant-product-country unit values feasible.
- **Main restricted sample.** The analysis focuses on textile and clothing HS products that are **quota-bound in one destination but quota-free in another**. Of 547 MFA products, removing the 188 products that were quota-bound in all three destinations leaves **359 HS categories** in the core sample.

2.2 Additional productivity evidence and timing

- **NBS manufacturing survey.** The paper also uses China’s annual industrial survey to compare exporter productivity by ownership inside textile and clothing industries.
- **Measured TFP.** The paper cannot directly match customs exporters to NBS firms well enough to compute exporter-level TFP for the main design, but it can compare textile/clothing exporters by ownership type in the NBS.
- **Key productivity fact.** In 2005, average log TFP is about **1.57** for SOEs, **3.19** for domestic private firms, and **2.73** for foreign firms. This is strong indirect evidence that SOEs were much less productive than private exporters.
- **Timing.** The paper’s main treatment is the removal of **Phase IV** MFA quotas on January 1, 2005. A key institutional advantage is that the list of Phase IV products was fixed back in 1995, so quota status in 2005 was not chosen in response to current shocks.

2.3 Constructed margins and empirical outcomes

- **Intensive margin.** Incumbents are firm-product-country triples observed in both 2004 and 2005.
- **Extensive margin.** Entrants are firm-product-country triples present in 2005 but not 2004; exiters are present in 2004 but not 2005.
- **Entry decomposition.** Entrants are further split into **new exporters** and **adders**. New exporters did not export anything in 2004; adders already exported some other products but added a quota-bound product in 2005.
- **Market shares.** The paper studies quantity-based rather than value-based market shares because prices move sharply after quota removal.
- **Prices and quality-adjusted prices.** The paper studies both raw unit values and a demand-based quality adjustment to make sure its price results are not simply about quality downgrading.

3 Models and empirical specifications (all equations + interpretation)

3.1 Auction-allocation model of quota access

The paper uses a simple heterogeneous-firm export model to generate predictions under an efficient quota allocation mechanism. Consumers in destination market c have CES preferences:

$$U = \left(\int_{\zeta \in \Omega} q_c(\zeta)^{(\sigma-1)/\sigma} d\zeta \right)^{\sigma/(\sigma-1)}.$$

Demand is therefore

$$q_c(\zeta) = p_c(\zeta)^{-\sigma} P_c^{\sigma-1} Y_c.$$

- **Interpretation.** This is the standard Melitz-style CES demand system.
- **Key friction.** Firms must pay a **per-unit license fee** a_{oc} to export under the quota. That fee clears the market for quota licenses.

Firm productivity is ϕ , and the delivered price in export market c is

$$p_{oc}(\phi, a_{oc}) = \frac{\sigma}{\sigma-1} \omega_o \left(\frac{\tau_{oc}}{\phi} + a_{oc} \right).$$

- **Interpretation.** The additive quota fee is the crucial term. Because it is per unit, it acts like a bigger percentage distortion on firms with lower marginal cost and lower pre-quota prices.
- **Economic intuition.** High-productivity firms are better able to pay the fee and remain exporters, but their quantities are disproportionately taxed by it. So once the fee disappears, the model predicts that the strongest growth should come from the most productive incumbents.

The productivity cutoff for exporting is

$$\phi_{oc}^* = \left[\left(\frac{\sigma-1}{\sigma} \right)^{\sigma/(1-\sigma)} \left(\frac{\omega_o f_{oc}}{Y_c} \right)^{1/(1-\sigma)} \frac{P_c}{\omega_o \tau_{oc}} - \frac{a_{oc}}{\tau_{oc}} \right]^{-1}.$$

- **Interpretation.** A higher quota license fee raises the productivity cutoff for exporting.
- **Prediction.** When quotas are relaxed or removed, lower-productivity firms can enter, but among incumbents market share should reallocate toward the largest and lowest-price firms.

3.2 Testable predictions of the efficient-allocation benchmark

Under auction allocation, quota removal implies three benchmark responses:

- the export quantities of the most productive incumbents should rise disproportionately;
- some lower-productivity firms may enter because they no longer need to pay the license fee;
- incumbents' prices should fall when the license fee disappears, but entry by relatively high-price firms should partly offset that decline.

- **Bottom line.** Under efficient allocation, **incumbents should dominate the adjustment.** The whole empirical case in the paper is that the data say the opposite.

3.3 Quality caveat

The paper explicitly notes a potential caveat: low prices may reflect low quality rather than high productivity. If more productive firms sell *higher-quality* goods at higher prices, then the low-price/high-productivity logic of the model could break down.

- **Implication.** In that case, entry by low-price firms after quota removal could be low-quality entry rather than high-productivity entry.
- **Response in the paper.** The authors therefore build a separate **quality-adjusted price** exercise before drawing conclusions from price changes.

3.4 Difference-in-differences identification design

The main empirical design compares quota-bound and quota-free exports of the *same* product across destinations before and after 2005:

$$\Delta Y_{hct} = \alpha_0 + \alpha_1 1\{t = 2005\} + \alpha_2 1\{hc \in \text{Quota-Bound}\} + \alpha_3 1\{t = 2005\} \times 1\{hc \in \text{Quota-Bound}\} + \varepsilon_{hct}.$$

- h indexes 8-digit HS product, c destination, and t year.
- The coefficient α_3 is the main **difference-in-differences estimate**.
- **Interpretation.** It measures whether outcomes for quota-bound product-country pairs changed disproportionately in 2005 relative to quota-free pairs.
- **Why the design is attractive.** A product like men’s cotton pajamas can be quota-bound in the United States and Canada but quota-free in the EU. That comparison differences out product-specific Chinese supply changes and broader demand changes.
- **Robustness.** The paper also estimates specifications with product-country fixed effects and placebo tests using 2003–2004 versus 2002–2003.

3.5 Market-share decomposition

Let Θ_{hct}^m be the quantity market share of margin m in product-country pair hc at date t :

$$\Theta_{hct}^m = \frac{\sum_{f \in m} q_{fhct}^m}{\sum_m \sum_f q_{fhct}^m}.$$

The paper runs the difference-in-differences specification with $\Delta Y_{hct} = \Theta_{hct}^m$ for incumbents, adders, new exporters, and exiters.

- **Interpretation.** This tells us whether adjustment happens on the intensive margin or on the extensive margin.
- **What efficient allocation predicts.** Incumbents should account for the bulk of the action.
- **What the paper finds.** Net entry does.

3.6 Price decomposition

The paper decomposes the change in average log export prices following Griliches and Regev (1995):

$$\Delta \bar{P}_{hct} = \frac{1}{\bar{P}_{hct-1}} \left[\sum_{f \in I} \bar{\theta}_{fhc} (\ln p_{fhct} - \ln p_{fhct-1}) + \sum_{f \in I} (\theta_{fhct} - \theta_{fhct-1}) (\bar{p}_{fhc} - \bar{P}_{hc}) \right] \quad (1)$$

$$+ \frac{1}{\bar{P}_{hct-1}} \left[\sum_{f \in N} \theta_{fhct} (\ln p_{fhct} - \bar{P}_{hc}) \right] - \frac{1}{\bar{P}_{hct-1}} \left[\sum_{f \in X} \theta_{fhct-1} (\ln p_{fhct-1} - \bar{P}_{hc}) \right]. \quad (2)$$

Here I denotes incumbents, N entrants, and X exiters.

- **Incumbent within component.** Price changes of incumbents holding their market shares fixed.
- **Incumbent across component.** Reallocation of market share toward higher- or lower-price incumbents.
- **Entry and exit components.** These capture whether entrants are cheaper than average and whether exiters are more expensive than average.
- **Interpretation.** If low-price entrants and high-price exiters dominate, then the price decline is an extensive-margin phenomenon rather than an incumbent phenomenon.

3.7 Quality-adjusted price specification

To purge price changes of quality differences, the paper estimates quality from a CES demand system with quality shifters:

$$\ln q_{fhct} + \sigma \ln p_{fhct} = \alpha_h + \alpha_{ct} + \varepsilon_{fhct}.$$

With $\sigma = 4$, estimated quality is the regression residual scaled by $1/(\sigma - 1)$, and quality-adjusted prices are

$$\ln p_{fhct} - \ln \hat{\lambda}_{fhct}.$$

- **Interpretation.** Conditional on price, a higher quantity is interpreted as higher quality.
- **Why it matters.** If post-2005 entry is just low-quality entry, then raw-price declines would overstate productivity gains. The paper shows the extensive-margin results survive this correction.

3.8 Back-of-the-envelope productivity calculation

Using ownership-specific TFP from the NBS and market-share changes from the customs data, the paper constructs a coarse estimate of the productivity gain from quota removal.

- **Result.** The observed 2005 market-share reallocation implies an increase in exporters' productivity of about **21.3%**.
- **Interpretation.** This is only a rough calculation because it uses average productivity by ownership type rather than firm-level productivity, but it shows the reallocation is quantitatively large.

3.9 Counterfactual decomposition: no quotas, auction allocation, political allocation

The paper then calibrates a quantitative model to decompose the total productivity gain from quota removal.

- **No-quota allocation.** This is the benchmark frictionless trade allocation once the quota is gone.
- **Auction allocation.** Quotas are still present, but licenses are assigned efficiently through a common market-clearing quota price.
- **Political allocation.** Quotas are assigned according to a separate “political connections” draw κ rather than productivity ϕ .

The counterfactual is disciplined by matching the empirical role of the extensive margin in price declines. In the political-allocation scenario, the rank correlation between productivity and political connection is set to

$$\rho = 0.15,$$

which makes the model reproduce the observed **68%** contribution of the extensive margin to the quality-adjusted price decline.

Key productivity numbers in the three scenarios are:

Political allocation: 1.51, Auction allocation: 3.43, No quota: 4.21.

- **Interpretation.** Moving from political to auction allocation eliminates the misallocation built into the licensing regime; moving from auction allocation to no quotas removes the trade barrier itself.
- **Main decomposition.** The contribution of institutional reform is

$$\frac{3.43 - 1.51}{4.21 - 1.51} \approx 0.71,$$

so about **71%** of the total productivity gain comes from eliminating quota misallocation, and 29% comes from removing the quota itself.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the empirical design credible

- **Destination-specific quota variation within the same product.** The same HS product can be quota-bound in one market and quota-free in another, which sharply improves comparability.
- **Phase IV predetermined.** The final-phase quota products were chosen in 1995, well before the 2005 removal, so treatment status is plausibly not driven by contemporaneous shocks.
- **Post-2005 SOE balance check.** The SOE share gap between quota-bound and quota-free exports disappears in 2005, which supports the idea that quota status was the relevant difference before removal.

- **Product-country fixed effects.** These absorb persistent heterogeneity across product-destination cells.
- **Placebo test.** There is no similar pre-trend in 2003–2004 versus 2002–2003.

4.2 What the paper identifies

- The paper does *not* identify the exact administrative objective function of the Chinese government.
- Instead, it identifies whether the allocation of quotas was **consistent with productivity-based allocation**.
- The answer is no: if quotas had been allocated according to productivity, incumbents would dominate the post-reform expansion and price declines. The data show the reverse.

4.3 Threats and how the paper addresses them

- **Quality differences.** Addressed by the quality-adjusted price exercise.
- **Capacity constraints of incumbents.** The paper argues these are unlikely to explain the results, because quota removal dates were known roughly a decade in advance and adding a new product-country line is harder than expanding an existing one.
- **Subcontracting / black-market quota trading.** The authors acknowledge that if subcontractors were hidden before 2005 and appear officially only after quota removal, the extensive margin could be overstated. But the counterfactual sensitivity analysis shows that even substantial overstatement would still leave institutional reform as the majority of total gains.

4.4 Relation to the literature

- **Misallocation literature.** The paper extends the focus from misallocation among incumbents to misallocation on the extensive margin, where efficient entrants are blocked.
- **Trade liberalization literature.** It argues that measured gains from trade can exceed textbook predictions when liberalization removes not only a tariff or quota but also the distorted institution that administered it.
- **Political economy.** The paper is also about embedded institutional reform: externally mandated quota removal can dismantle domestic arrangements that would otherwise be politically difficult to reform.

5 Tables and figures

5.1 Table 1: Export value and number of exporters

Read:

- Quota-bound exports surge after quota removal. Export value for quota-bound goods rises from \$8.9 billion in 2004 to \$19.6 billion in 2005, a **119%** one-year increase.

- Quota-free goods also grow, but much less: from \$6.3 billion to \$8.2 billion, a **29%** increase.
- The number of exporters also jumps much more for quota-bound goods: from 9,523 to 18,628, a **96%** increase between 2004 and 2005, versus 47% for quota-free goods.
- This table establishes that the quotas were genuinely binding and that 2005 is a major treatment shock.

TABLE 1—EXPORT VALUE AND NUMBER OF EXPORTERS

	Export value (\$billion)	
	Quota free	Quota bound
2000	2.4	4.8
2001	3.2	6.2
2002	3.8	6.5
2003	5.3	7.9
2004	6.3	8.9
2005	8.2	19.6
Percent growth 2000–2005	248	307
Annual percent growth 2000–2004	28	17
Annual percent growth 2004–2005	29	119

	Number of exporting firms	
	Quota free	Quota bound
2000	3,525	3,536
2001	4,518	4,253
2002	6,469	5,911
2003	8,501	7,793
2004	10,718	9,523
2005	15,756	18,628
Percent growth 2000–2005	347	427
Annual percent growth 2000–2004	32	28
Annual percent growth 2004–2005	47	96

Notes: Panels report annual export value and number of exporters by classification. Exports in the 547 HS products are classified as quota free and quota bound if they are or are not subject to quotas in 2004, respectively.

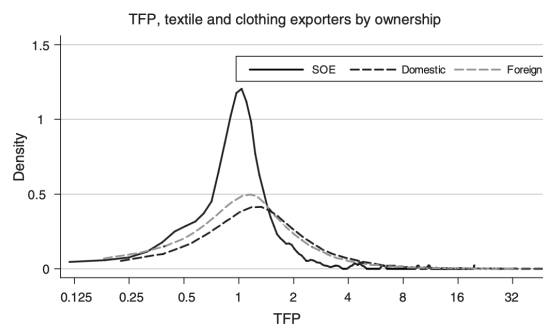


FIGURE 1. TEXTILE AND APPAREL PRODUCERS' TFP, 2005

Notes: First and ninety-ninth percentiles are dropped from each distribution. Collective firms are excluded.

5.2 Figure 1: TFP of textile and clothing exporters by ownership

Read:

- The SOE productivity distribution lies clearly to the left of the domestic and foreign private distributions.
- Average log TFP is about **1.57** for SOEs, **3.19** for domestic firms, and **2.73** for foreign firms.
- This figure is central because the customs data do not directly reveal firm TFP for the main design. The ownership composition of reallocation is therefore informative about productivity reallocation.
- The message is simple: reallocation away from SOEs and toward private firms is very likely to be productivity improving.

5.3 Table 2: SOE market share by year

Read:

- Before quota removal, quota-bound exports have a significantly larger SOE share than quota-free exports: about **0.084**, **0.089**, and **0.090** in 2002, 2003, and 2004.
- In 2005, that gap becomes **-0.020** and is statistically insignificant.
- This is an important validation check for the difference-in-differences design.

- The most natural reading is that the pre-2005 excess SOE share was created by the quota allocation regime itself.

TABLE 2—SOE MARKET SHARE, BY YEAR

	2002 (1)	2003 (2)	2004 (3)	2005 (4)
Quota-Bound _{it}	0.084*** 0.019	0.089*** 0.019	0.090*** 0.020	−0.020 0.017
Constant	0.675*** 0.013	0.596*** 0.013	0.534*** 0.014	0.421*** 0.012
Observations	932	943	949	1,016
R ²	0.02	0.02	0.02	0.00

Notes: Table reports regressions of the share of SOE exports in a HS-country pair in the year noted by each column. This table uses the restricted sample of HS codes defined in Section IIIB. The Quota-Bound dummy is a variable that equals one if the export was subject to a quota in 2004, and the coefficient is the difference in SOE shares between quota-bound and quota-free exports.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 3—DECOMPOSITION OF MARKET SHARE CHANGES

Difference-in-differences (Quota-Bound versus Quota-Free, 2004–2005 versus 2003–2004)				
Margin	All	SOE	Domestic	Foreign
Incumbents	−0.122***	−0.106***	−0.013	−0.003
Net entry				
Adders	0.116***	−0.011	0.071***	0.056***
New exporters	0.037***	−0.003	0.035***	0.005
Exiters	−0.031	−0.027	−0.001	−0.003
Total net entry	0.122***	−0.041	0.105***	0.058***
Total	0.000	−0.147***	0.092***	0.055**
Difference-in-differences, country-product FEs (Quota-Bound versus Quota-Free, 2004–2005 versus 2003–2004)				
Margin	All	SOE	Domestic	Foreign
Incumbents	−0.124***	−0.107***	−0.014	−0.003
Net entry				
Adders	0.116***	−0.010	0.073***	0.053***
New exporters	0.038***	−0.003	0.036***	0.005
Exiters	−0.030	−0.021	0.000	−0.009
Total net entry	0.124***	−0.034	0.109***	0.049
Total	0.000	−0.141***	0.095**	0.046
Prereform difference-in-differences (Quota-Bound versus Quota-Free, 2003–2004 versus 2002–2003)				
Margin	All	SOE	Domestic	Foreign
Incumbents	−0.016	−0.001	0.006	−0.021**
Net entry				
Adders	0.017	0.002	−0.005	0.020
New exporters	−0.024**	−0.010*	−0.013	−0.001
Exiters	0.024	0.024	0.015	−0.015
Total net entry	0.016	0.016	−0.003	0.003
Total	0.000	0.015	0.002	−0.018

Notes: Table reports the coefficient α^3 in equation (3), where the dependent variable in each regression is the change in quantity-based market share for each margin-firm ownership type. The regression is run on the restricted sample of HS codes defined in Section IIIB. The top panel reports the relative change in 2004 to 2005 versus 2003 to 2004 in quantity-based market share between Quota-Bound versus Quota-Free exports, by margin of adjustment and firm ownership type. The middle panel is analogous to the top panel, but includes country-product pair fixed effects. The bottom panel runs the regression on the pre-reform sample from 2003 to 2004 versus 2002 to 2003. In all panels, rows 2 to 4 sum to row 5, rows 1 and 5 sum to row 6, and the first column is sum of the remaining columns. Standard errors are clustered by HS product.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.4 Table 3: Decomposition of market share changes

Read:

- Incumbents lose **12.2 percentage points** of market share in quota-bound product-country pairs after quota removal, while total net entry gains exactly **12.2 points**.
- Most of the entry gain comes from **adders** (11.6 points), not entirely new exporters (3.7 points); exiters reduce the net effect by 3.1 points.
- SOEs lose **14.7 percentage points** of market share overall, while domestic and foreign private firms gain **9.2** and **5.5** points, respectively.
- This table is the core rejection of the auction-allocation benchmark: adjustment is dominated by net entry and by private firms, not by the most productive incumbents.

5.5 Figures 2 and 3: Distribution of export prices by margin

Read:

- In quota-bound products after reform, entrants have lower prices than both incumbents and exiters.
- On average, entrants' prices are about **0.26 log points** lower than incumbents' prices and about **0.21 log points** lower than exiters' prices.

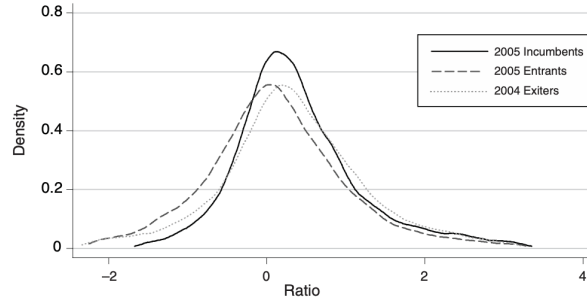


FIGURE 2. DISTRIBUTION OF QUOTA-BOUND EXPORT PRICES, BY MARGIN

Note: First and ninety-ninth percentiles are dropped from each distribution.

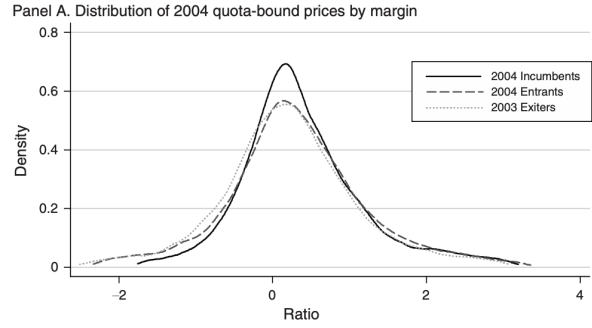


FIGURE 3. DISTRIBUTION OF EXPORT PRICES IN COMPARATOR GROUPS

Note: First and ninety-ninth percentiles are dropped from each distribution.

- The analogous comparator distributions do not show the same strong ordering in the prereform year or in quota-free products.
- This visual evidence supports the idea that post-2005 net entry is disproportionately associated with low-price producers.

5.6 Table 4: Decomposition of export price changes

Read:

- Quota removal is associated with a **0.206 log point** relative decline in raw export prices.
- The extensive margin explains about **58%** of that total price decline.
- Among incumbents, most of the decline comes from the **across** component, meaning high-price incumbents lose market share.
- By ownership, SOEs and domestic private firms explain almost all of the overall decline; for domestic firms, the extensive margin is especially important.

5.7 Table 5: Decomposition of quality-adjusted export price changes

Read:

- Even after adjusting for quality, quota removal is associated with a **0.166 log point** relative price decline.

TABLE 4—DECOMPOSITION OF EXPORT PRICE CHANGES

Margin	Difference-in-differences (Quota-Bound versus Quota-Free, 2004–2005 versus 2003–2004)			
	All	SOE	Domestic	Foreign
Incumbents (I)				
Within	−0.037***	−0.023**	−0.009	−0.005
Across	−0.049***	−0.028**	−0.012**	−0.008
Entrant (N)	−0.069***	−0.021	−0.050***	0.002
Exiters (X)	0.051**	0.022	0.028**	0.000
Net entry (N − X)	−0.120**	−0.044	−0.078***	0.002
Total	−0.206***	−0.095**	−0.100***	−0.012
Extensive share	0.582	0.459	0.786	−0.167
Difference-in-differences, country-product FEs (Quota-Bound versus Quota-Free, 2004–2005 versus 2003–2004)				
Margin	All	SOE	Domestic	Foreign
Incumbents (I)				
Within	−0.041**	−0.026*	−0.010	−0.005
Across	−0.049**	−0.028	−0.012*	−0.009
Entrant (N)	−0.075**	−0.024	−0.046**	−0.005
Exiters (X)	0.055	0.027	0.028	−0.001
Net entry (N − X)	−0.130*	−0.051	−0.074**	−0.005
Total	−0.221***	−0.106*	−0.096***	−0.019
Extensive share	0.589	0.487	0.769	0.256
Prereform difference-in-differences (Quota-Bound versus Quota-Free, 2003–2004 versus 2002–2003)				
Margin	All	SOE	Domestic	Foreign
Incumbents (I)				
Within	−0.018	−0.014	−0.004	0.000
Across	0.007	0.007	0.003	−0.003
Entrant (N)	−0.019	−0.003	0.005	−0.021*
Exiters (X)	0.027	0.048**	−0.013	−0.008
Net entry (N − X)	−0.046	−0.051	0.018	−0.013
Total	−0.058	−0.058	0.017	−0.017
Extensive share	0.801	0.878	1.068	0.804

Notes: Table reports the coefficient α' in equation (3), where the dependent variable in each regression is the change in prices for each margin-firm ownership type. Equation (6) explains how each price margin is computed. The regression is run on the restricted sample of HS codes defined in Section IIIB. The top panel reports the relative change in 2004 to 2005 versus 2003 to 2004 between Quota-Bound versus Quota-Free exports, by margin of adjustment and firm ownership type. The middle panel is analogous to the left panel but includes country-product pair fixed effects. The bottom panel runs the regression on the prereform sample from 2003 to 2004 versus 2002 to 2003. In all panels, rows 2 to 4 sum to row 5, rows 1 and 5 sum to row 6, and the first column is the sum of the remaining columns. Standard errors are clustered by HS product.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 5—DECOMPOSITION OF QUALITY-ADJUSTED EXPORT PRICE CHANGES

Margin	Difference-in-differences (Quota-Bound versus Quota-Free, 2004–2005 versus 2003–2004)			
	All	SOE	Domestic	Foreign
Incumbents (I)				
Within	−0.055***	−0.026**	−0.011	−0.018**
Across	0.001	−0.005	−0.002	0.007
Entrant (N)	−0.072***	−0.026**	−0.029***	−0.018***
Exiters (X)	0.040**	0.032**	0.007	0.000
Net entry (N − X)	−0.112***	−0.058**	−0.036**	−0.018*
Total	−0.166***	−0.088***	−0.049***	−0.028**
Extensive share	0.675	0.653	0.736	0.639
Difference-in-differences, country-product FEs (Quota-Bound versus Quota-Free, 2004–2005 versus 2003–2004)				
Margin	All	SOE	Domestic	Foreign
Incumbents (I)				
Within	−0.054**	−0.025	−0.011	−0.018*
Across	0.001	−0.004	−0.002	0.007
Entrant (N)	−0.080***	−0.027	−0.031**	−0.022***
Exiters (X)	0.048**	0.036*	0.009	0.003
Net entry (N − X)	−0.129***	−0.064*	−0.040*	−0.025*
Total	−0.182***	−0.093**	−0.054**	−0.028**
Extensive share	0.708	0.688	0.750	0.698
Prereform difference-in-differences (Quota-Bound versus Quota-Free, 2003–2004 versus 2002–2003)				
Margin	All	SOE	Domestic	Foreign
Incumbents (I)				
Within	0.018	0.013	−0.005	0.010
Across	−0.018*	−0.010	−0.001	−0.006
Entrant (N)	0.004	0.005	−0.005	0.004
Exiters (X)	−0.007	−0.004	−0.002	−0.001
Net entry (N − X)	0.011	0.009	−0.003	0.005
Total	0.012	0.012	−0.008	0.009
Extensive share	0.932	0.768	0.320	0.581

Notes: Table reports the coefficient α_3 in equation (3), where the dependent variable in each regression is the change in quality for each margin-firm ownership type. Section IVC describes how quality is measured, and equation (6) explains how each margin is computed, using quality instead of prices. The regression is run on the restricted sample of HS codes defined in Section IIIB. The top panel reports the relative change in 2004 to 2005 versus 2003 to 2004 between Quota-Bound versus Quota-Free exports, by margin of adjustment and firm ownership type. The middle panel is analogous to the top panel, but includes country-product pair fixed effects. The bottom panel runs the regression on the prereform sample from 2003 to 2004 versus 2002 to 2003. In all panels, rows 2 to 4 sum to row 5, rows 1 and 5 sum to row 6, and the first column is the sum of the remaining columns. Standard errors are clustered by HS product.

***Significant at the 1 percent level. **Significant at the 5 percent level. *Significant at the 10 percent level.

- The extensive margin explains about **67.5%** of that decline, even more than in raw prices.
- So the key result survives the quality correction: the price decline is still mainly about entry and exit, not incumbent price changes.
- The paper uses this **68%** extensive-margin contribution to pin down the degree of misallocation in the political-allocation counterfactual.

5.8 Figure 4: Sensitivity of inferred misallocation to the extensive margin

Read:

- The left panel shows that as the extensive-margin contribution rises, the implied productivity of the political allocation falls relative to the auction allocation.
- The right panel shows that the implied contribution of institutional reform to total productivity gains rises quickly with the extensive margin.
- At the observed 68% extensive-margin contribution, institutional reform accounts for about **71%** of the overall gain.
- The authors also note that even if the extensive margin were overstated by a factor of three, institutional reform would still account for the majority of the overall productivity gain.

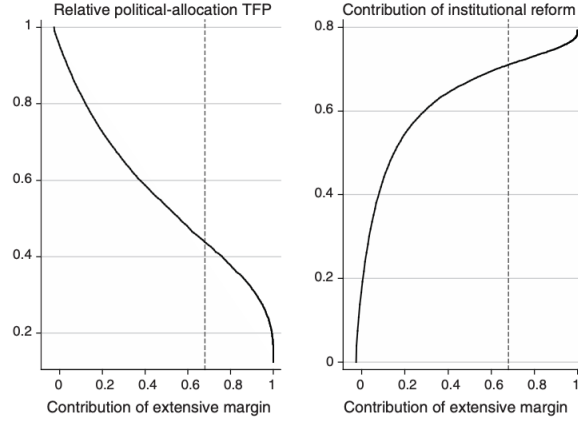


FIGURE 4. SENSITIVITY OF INFERRED MISALLOCATION TO EXTENSIVE MARGIN'S CONTRIBUTION TO PRICE DECLINES AFTER QUOTAS ARE REMOVED

Notes: Left panel displays weighted-average firm TFP under political allocation as a share of weighted-average firm TFP under auction allocation. Right panel traces out the share of overall productivity growth accounted for by institutional reform. Both quantities are plotted against the extensive margin's contribution to the overall price decline when quotas are removed. Dashed vertical lines indicate the observed contribution of the extensive margin from Table 5. Contribution of extensive margin is negative if quota removal induces low-productivity (high-price) firms to enter.

6 Short summary

- **Conclusion.** Chinese quota removal in 2005 did more than eliminate a trade barrier. It also dismantled a quota allocation system that had favored relatively inefficient firms.
- **Identification insight.** The paper identifies misallocation not by observing quotas directly, but by asking whether post-reform adjustment matches the predictions of an efficient allocation benchmark.
- **Main empirical lesson.** It does not. Export growth and price declines are driven by private entry and the decline of SOE incumbents, implying that the quota regime blocked productive firms from exporting.
- **Main quantitative lesson.** Most of the productivity gain from quota removal appears to come from eliminating the embedded institutional distortion rather than from eliminating the quota itself.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the Chinese export licensing regime under the MFA was not simply a neutral administrative device for implementing quotas.
- Instead, it was an allocation mechanism that appears to have systematically deviated from firm productivity, favoring less productive incumbents and restricting more productive private firms.
- The empirical signature of this distortion is the dominance of net entry and the movement of market share away from SOEs after quota removal.

- In the quantitative decomposition, this institutional distortion is responsible for the majority of total productivity gains from liberalization.

7.2 Economic intuition

- A quota is already distortionary because it limits total trade. But if the rights to export under the quota are handed to the wrong firms, there is a second layer of inefficiency.
- In the efficient benchmark, the most productive firms should be the ones exporting under a quota, because they can best absorb the per-unit restriction.
- If instead quotas are handed to politically connected or otherwise favored low-productivity firms, trade is distorted twice: first by the quota, and second by who gets to use it.
- Liberalization then creates two gains: a standard gain from removing the barrier and an additional reallocation gain from shutting down the distorted institution that allocated it.

7.3 Takeaway

This paper is best read as a misallocation paper hidden inside a trade-liberalization episode. The main message is not merely that quotas reduced trade. It is that the institution created to manage quotas in China appears to have blocked efficient private exporters and protected inefficient incumbents, especially SOEs. When the quota system disappeared in 2005, market share and low prices came disproportionately from entrants, not from incumbent expansion, which is exactly the opposite of what an efficient allocation of quota rights would predict. That is why the gains from liberalization are so large in the paper: removing quotas also removed the institution that was misallocating them.